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## JEE MAIN PHYSICS DPP: LAW OF CONSERVATION OF LINEAR MOMENTUM

Topic: Conservation of Linear Momentum, Derivation, Recoil Dynamics & Explosions | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** The law of conservation of linear momentum for an isolated two-body system is derived by combining:
- Newton's first and third laws
  - Newton's first and second laws
  - Newton's second and third laws
  - Law of conservation of energy and Newton's first law
- Q2.** A stationary nucleus of mass number  $A$  decays by emitting an alpha particle ( ${}^4\text{He}$ ) with speed  $v_\alpha$ . The recoil speed of the daughter nucleus is:
- $\frac{4v_\alpha}{A-4}$
  - $\frac{4v_\alpha}{A}$
  - $\frac{Av_\alpha}{A-4}$
  - $\frac{(A-4)v_\alpha}{4}$
- Q3.** A bomb of mass 9 kg initially at rest explodes into two fragments of masses 3 kg and 6 kg. If the 3 kg fragment flies off with kinetic energy 216 J, the total energy released in the explosion is:
- 216 J
  - 432 J
  - 648 J
  - 324 J
- Q4.** A particle of mass  $m$  moving with velocity  $\vec{v} = v_0 \hat{i}$  collides with a stationary particle of mass  $2m$  and sticks to it. The common velocity of the composite body after collision is:
- $\frac{v_0}{2} \hat{i}$
  - $\frac{v_0}{3} \hat{i}$
  - $\frac{2v_0}{3} \hat{i}$
  - $\frac{v_0}{4} \hat{i}$
- Q5.** A shell of mass  $m$  is projected from the ground with speed  $u$  at an angle  $\theta$  to the horizontal. At the highest point of its trajectory, it explodes into two equal fragments. If one fragment retraces its path back to the launch point, the speed of the other fragment just after explosion is:
- $3u \cos \theta$
  - $2u \cos \theta$
  - $u \cos \theta$
  - $4u \cos \theta$
- Q6.** A person of mass  $M = 60$  kg standing at one end of a stationary boat of mass  $m = 140$  kg and length  $L = 5$  m on calm water walks to the other end of the boat. Neglecting water resistance, the displacement of the boat relative to the water is:
- 2.0 m
  - 3.5 m
  - 1.5 m
  - 2.5 m
- Q7.** A stationary body explodes into three fragments of masses in the ratio 1 : 1 : 2. The two fragments of equal mass fly off along mutually perpendicular directions with speed 30 m/s each. The speed of the third heavier fragment is:
- $30\sqrt{2}$  m/s
  - $15\sqrt{2}$  m/s
  - 15 m/s
  - 30 m/s
- Q8.** Under which of the following conditions is the linear momentum of a system of particles strictly conserved along the  $x$ -axis?
- When total mechanical energy is conserved
  - When the net force acting on each particle is zero
  - When internal forces between particles cancel out
  - When the  $x$ -component of the net external force is zero ( $\sum F_{\text{ext},x} = 0$ )
- Q9.** A gun of mass  $M = 4$  kg fires a bullet of mass  $m = 50$  g with a velocity  $v = 400$  m/s. The ratio of the kinetic energy of the bullet to that of the gun is:
- 80 : 1
  - 1 : 80
  - 40 : 1
  - 160 : 1
- Q10.** A heavy trolley of mass  $M = 100$  kg is moving with a constant speed  $v_0 = 10$  m/s on frictionless rails. A boy of mass  $m = 50$  kg runs on the trolley with a speed of  $u = 3$  m/s relative to the trolley in the forward direction. The final speed of the trolley relative to the ground is:
- 10.0 m/s

- (b) 9.0 m/s  
(c) 8.0 m/s  
(d) 11.0 m/s
- Q11.** In a two-body interaction, particle 1 exerts force  $\vec{F}_{12}$  on particle 2 for time  $\Delta t$ . During the derivation of momentum conservation, which statement represents the core algebraic step?
- (a)  $\vec{F}_{12}\Delta t = \vec{0}$   
(b)  $\Delta\vec{p}_1 + \Delta\vec{p}_2 = \vec{F}_{\text{ext}}\Delta t$   
(c)  $\vec{F}_{12} = m_1\vec{a}_1$   
(d)  $\Delta\vec{p}_1 + \Delta\vec{p}_2 = (\vec{F}_{21} + \vec{F}_{12})\Delta t = \vec{0}$
- Q12.** A flatcar of mass  $M = 200$  kg is at rest on a frictionless track. Two men, each of mass  $m = 50$  kg, jump off the rear end of the car with horizontal velocity  $u = 4$  m/s relative to the car. The final speed of the flatcar if both men jump simultaneously is:
- (a) 1.50 m/s  
(b) 1.00 m/s  
(c) 1.33 m/s  
(d) 1.71 m/s
- Q13.** A rocket of mass  $m$  in deep space expels a mass  $\Delta m$  with speed  $u$  relative to the rocket during time  $\Delta t$ . The governing differential equation of motion obtained from momentum conservation is:
- (a)  $m \frac{dv}{dt} = u \frac{dm}{dt}$   
(b)  $m \frac{dv}{dt} = -u \frac{dm}{dt}$   
(c)  $\frac{d(mv)}{dt} = 0$   
(d)  $v \frac{dm}{dt} = -m \frac{dv}{dt}$
- Q14.** A ball of mass  $m_1 = 0.5$  kg moving at 6 m/s collides head-on with a stationary ball of mass  $m_2 = 1.0$  kg. If the collision is perfectly elastic, the velocity of the 0.5 kg ball after collision is:
- (a) -2.0 m/s  
(b) +2.0 m/s  
(c) -4.0 m/s  
(d) 0 m/s
- Q15.** A bullet of mass  $m = 10$  g moving horizontally with velocity  $v = 400$  m/s strikes a wooden block of mass  $M = 3.99$  kg suspended by a light vertical string of length  $L = 1$  m and gets embedded in it. The maximum vertical height  $h$  to which the combined block-bullet system rises is: (Take  $g = 10$  m/s<sup>2</sup>)
- (a) 0.10 m  
(b) 0.20 m  
(c) 0.50 m  
(d) 0.05 m
- Q16.** A piece of wood of mass  $M = 3$  kg is dropped from a tower. At  $t = 2$  s, a bullet of mass  $m = 1$  kg fired vertically upward with initial speed 100 m/s ( $v_b = 80$  m/s at  $t = 2$  s) collides and embeds in the wood. The velocity of the combined mass immediately after collision is: (Take  $g = 10$  m/s<sup>2</sup>)
- (a) -5 m/s  
(b) +10 m/s  
(c) +5 m/s  
(d) -10 m/s
- Q17.** If the net external force acting on a system of particles is zero ( $\sum \vec{F}_{\text{ext}} = \vec{0}$ ), which of the following remains strictly constant?
- (a) Total kinetic energy of the system  
(b) Velocity of the center of mass of the system  
(c) Individual velocities of all constituent particles  
(d) Potential energy of the system
- Q18.** A particle of mass  $m$  collides with a stationary particle of mass  $M$ . For a completely inelastic collision in one dimension, the fraction of initial kinetic energy lost ( $\frac{\Delta K}{K_i}$ ) is:
- (a)  $\frac{m}{m+M}$   
(b)  $\frac{M-m}{M+m}$   
(c)  $\frac{m^2}{(m+M)^2}$   
(d)  $\frac{M}{m+M}$
- Q19.** A projectile of mass  $M$  is fired from the ground with velocity  $v_0$  at angle  $\alpha$ . At the highest point of its flight, it splits into two fragments of masses  $M/3$  and  $2M/3$ . If the lighter fragment falls vertically downward with zero horizontal velocity, the distance from the launch point where the heavier fragment hits the ground is: ( $R$  is original range)
- (a)  $\frac{5}{4}R = 1.25R$   
(b)  $\frac{3}{2}R = 1.50R$   
(c)  $\frac{4}{3}R \approx 1.33R$   
(d)  $2R$
- Q20.** A block of mass  $m = 2$  kg is placed on a triangular wedge of mass  $M = 8$  kg resting on a smooth horizontal floor. The block is released from rest at height  $h = 1$  m on the smooth incline. When the block reaches the floor, the speed of the wedge is: (Take  $g = 10$  m/s<sup>2</sup>)
- (a) 2.0 m/s  
(b) 4.0 m/s  
(c) 1.0 m/s  
(d) 0.5 m/s

# ANSWER KEY & STEP-BY-STEP SOLUTIONS

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## Quick Answer Key

| Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans |
|------|-----|------|-----|------|-----|------|-----|
| 1    | (c) | 6    | (c) | 11   | (d) | 16   | (c) |
| 2    | (a) | 7    | (b) | 12   | (c) | 17   | (b) |
| 3    | (d) | 8    | (d) | 13   | (b) | 18   | (d) |
| 4    | (b) | 9    | (a) | 14   | (a) | 19   | (a) |
| 5    | (a) | 10   | (b) | 15   | (d) | 20   | (c) |

## Detailed Solutions

### Sol 1. Correct Option: (c)

By Newton's second law:

$$\vec{F}_{21} = \frac{d\vec{p}_1}{dt}, \quad \vec{F}_{12} = \frac{d\vec{p}_2}{dt}$$

By Newton's third law:  $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$ . Adding both equations:

$$\frac{d}{dt}(\vec{p}_1 + \vec{p}_2) = \vec{0} \implies \vec{p}_1 + \vec{p}_2 = \text{constant}$$

### Sol 2. Correct Option: (a)

Initial momentum of the parent nucleus is zero ( $P_i = 0$ ). By conservation of linear momentum:

$$(A - 4)v_D - 4v_\alpha = 0 \implies v_D = \frac{4v_\alpha}{A - 4}$$

### Sol 3. Correct Option: (d)

By momentum conservation ( $p_1 = p_2 = p$ ):

$$K = \frac{p^2}{2m} \implies \frac{K_2}{K_1} = \frac{m_1}{m_2} = \frac{3}{6} = \frac{1}{2}$$

Given  $K_1 = 216$  J (for 3 kg fragment):

$$K_2 = \frac{216}{2} = 108 \text{ J}$$

Total explosive energy released:

$$E_{\text{total}} = K_1 + K_2 = 216 + 108 = 324 \text{ J}$$

### Sol 4. Correct Option: (b)

By linear momentum conservation:

$$mv_0\hat{i} + 2m(\vec{0}) = (m + 2m)\vec{v}_f$$

$$\vec{v}_f = \frac{mv_0}{3m}\hat{i} = \frac{v_0}{3}\hat{i}$$

### Sol 5. Correct Option: (a)

At the highest point, velocity before explosion is purely horizontal:  $\vec{v}_i = (u \cos \theta)\hat{i}$ . Total momentum before explosion:  $\vec{P}_i = m(u \cos \theta)\hat{i}$ . For the retracing fragment:  $\vec{v}_1 = -(u \cos \theta)\hat{i}$ . By momentum conservation:

$$m(u \cos \theta)\hat{i} = \frac{m}{2}(-u \cos \theta\hat{i}) + \frac{m}{2}\vec{v}_2$$

$$\vec{v}_2 = 3u \cos \theta\hat{i}$$

### Sol 6. Correct Option: (c)

Since no external horizontal force acts on the (man + boat) system, the center of mass remains stationary ( $\Delta x_{\text{cm}} = 0$ ). Let  $x$  be the backward shift of the boat:

$$M(L - x) - mx = 0 \implies x = \frac{ML}{M + m}$$

$$x = \frac{60 \times 5}{60 + 140} = \frac{300}{200} = 1.5 \text{ m}$$

### Sol 7. Correct Option: (b)

Let mass ratio be  $m : m : 2m$ . Momentum of first fragment:  $\vec{p}_1 = m(30)\hat{i}$ . Momentum of second fragment:  $\vec{p}_2 = m(30)\hat{j}$ . By momentum conservation ( $\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = \vec{0}$ ):

$$\vec{p}_3 = -(30m\hat{i} + 30m\hat{j})$$

Magnitude:

$$|\vec{p}_3| = \sqrt{(30m)^2 + (30m)^2} = 30\sqrt{2}m$$

Since mass of third fragment is  $2m$ :

$$(2m)v_3 = 30\sqrt{2}m \implies v_3 = \frac{30\sqrt{2}}{2} = 15\sqrt{2} \text{ m/s}$$

### Sol 8. Correct Option: (d)

Newton's second law for a multi-particle system gives  $\sum \vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt}$ . Component-wise, if  $\sum F_{\text{ext},x} = 0$ , then  $\frac{dP_x}{dt} = 0 \implies P_x = \text{constant}$ .

### Sol 9. Correct Option: (a)

Since  $|p_{\text{bullet}}| = |p_{\text{gun}}| = p$ :

$$K = \frac{p^2}{2m} \implies \frac{K_{\text{bullet}}}{K_{\text{gun}}} = \frac{M_{\text{gun}}}{m_{\text{bullet}}} = \frac{4 \text{ kg}}{0.05 \text{ kg}} = 80$$

Ratio is 80 : 1.

### Sol 10. Correct Option: (b)

Initial momentum:  $P_i = (M + m)v_0 = (100 + 50)(10) = 1500 \text{ kg} \cdot \text{m/s}$ . Let final trolley velocity relative to ground be  $v_T$ . Boy's velocity relative to ground:  $v_{\text{boy}} = v_T + u = v_T + 3$ . Conservation of momentum:

$$Mv_T + m(v_T + 3) = 1500$$

$$100v_T + 50(v_T + 3) = 1500 \implies 150v_T = 1350$$

$$v_T = \frac{1350}{150} = 9.0 \text{ m/s}$$

**Sol 11. Correct Option: (d)**

Newton's second law gives  $\vec{F}_{21}\Delta t = \Delta\vec{p}_1$  and  $\vec{F}_{12}\Delta t = \Delta\vec{p}_2$ . Using  $\vec{F}_{12} = -\vec{F}_{21}$ , adding yields:

$$\Delta\vec{p}_1 + \Delta\vec{p}_2 = (\vec{F}_{21} + \vec{F}_{12})\Delta t = \vec{0}$$

**Sol 12. Correct Option: (c)**

Initial momentum of car + men is zero ( $P_i = 0$ ). Let  $v_c$  be the forward velocity of the flatcar. Velocity of each man relative to ground:  $v_m = -(u - v_c)$ . Conservation of momentum:

$$Mv_c + 2m[-(u - v_c)] = 0$$

$$200v_c - 100(4 - v_c) = 0$$

$$300v_c = 400 \implies v_c \approx 1.33 \text{ m/s}$$

**Sol 13. Correct Option: (b)**

By momentum conservation for variable mass:

$$mv = (m + dm)(v + dv) - dm(v - u)$$

Expanding and dropping second-order differentials ( $dm \cdot dv$ ):

$$mdv = -udm \implies m \frac{dv}{dt} = -u \frac{dm}{dt}$$

**Sol 14. Correct Option: (a)**

For elastic head-on collision with  $u_2 = 0$ :

$$v_1 = \left( \frac{m_1 - m_2}{m_1 + m_2} \right) u_1 = \left( \frac{0.5 - 1.0}{0.5 + 1.0} \right) (6)$$

$$v_1 = \left( -\frac{0.5}{1.5} \right) (6) = -\frac{1}{3}(6) = -2.0 \text{ m/s}$$

**Sol 15. Correct Option: (d)**

1. Momentum conservation during embedding:

$$mv = (M + m)V_{\text{common}}$$

$$4 = 4V_{\text{common}} \implies V_{\text{common}} = 1.0 \text{ m/s}$$

$$V_{\text{common}} = 1.0 \text{ m/s}$$

2. Energy conservation during upward swing:

$$\frac{1}{2}(M + m)V_{\text{common}}^2 = (M + m)gh$$

$$h = \frac{V_{\text{common}}^2}{2g} = \frac{1^2}{2(10)} = 0.05 \text{ m}$$

**Sol 16. Correct Option: (c)**

Velocity of wood:  $v_w = -gt = -20 \text{ m/s}$ .

Velocity of bullet:  $v_b = 100 - 10(2) = +80 \text{ m/s}$ .

By conservation of momentum:

$$Mv_w + mv_b = (M + m)V_f$$

$$3(-20) + 1(80) = (3 + 1)V_f \implies V_f = +5 \text{ m/s}$$

**Sol 17. Correct Option: (b)**

When  $\sum \vec{F}_{\text{ext}} = \vec{0}$ , the center of mass acceleration is zero ( $\vec{a}_{\text{cm}} = \vec{0}$ ). Thus, velocity of center of mass  $\vec{v}_{\text{cm}}$  remains strictly constant.

**Sol 18. Correct Option: (d)**

$$\text{Common velocity: } V = \frac{mv}{m + M}.$$

Fraction of energy lost:

$$\frac{\Delta K}{K_i} = 1 - \frac{K_f}{K_i} = 1 - \frac{m}{m + M} = \frac{M}{m + M}$$

**Sol 19. Correct Option: (a)**

At top, original distance covered is  $R/2$ . Horizontal momentum before explosion:  $Mv_x$ . After explosion:  $Mv_x = \frac{M}{3}(0) + \frac{2M}{3}v_2 \implies v_2 = \frac{3}{2}v_x$ . Both pieces take remaining time  $T/2$  to land:

$$\Delta x_2 = \left( \frac{3}{2}v_x \right) \left( \frac{T}{2} \right) = \frac{3}{4}R$$

Landing point from launch:

$$X = \frac{R}{2} + \frac{3}{4}R = \frac{5}{4}R = 1.25R$$

**Sol 20. Correct Option: (c)**

Since no external horizontal force acts on (block + wedge):

$$MV_w = mv_{bx} \implies V_w = \frac{m}{M}v_{bx} = \frac{2}{8}v_{bx} = \frac{v_{bx}}{4}$$

Conservation of mechanical energy:

$$mgh = \frac{1}{2}MV_w^2 + \frac{1}{2}mv_b^2$$

For horizontal floor landing,  $v_b = v_{bx}$ :

$$2(10)(1) = \frac{1}{2}(8)V_w^2 + \frac{1}{2}(2)(4V_w)^2$$

$$20 = 4V_w^2 + 16V_w^2 = 20V_w^2 \implies V_w = 1.0 \text{ m/s}$$

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